

Multi-Faceted Interpolation

Classic OT

Sinkhorn Divergence

MMD

Computational
Complexity

$$\mathcal{O}(n^3 \log n)$$

$$\mathcal{O}(n^2/\varepsilon^2)$$

$$n^2$$

Statistical
Complexity

$$\mathcal{O}(n^{-\frac{1}{d}})$$

$$\mathcal{O}(\epsilon^{-\frac{d}{2}} n^{-\frac{1}{2}})$$

$$\mathcal{O}(n^{-\frac{1}{2}})$$

Coupling

Hard/sparse
coupling

Soft/dense
coupling

Naive (independent)
coupling

Geometry

Via cost

Via cost

Via kernel

Why geometry awareness matters

$$\text{TV}(\delta_{x_t}, \delta_x) = 1$$

$$\text{KL}(\delta_{x_t}, \delta_x) = \infty$$

$$W_p(\delta_{x_t}, \delta_x) = d(x_t, x)$$

